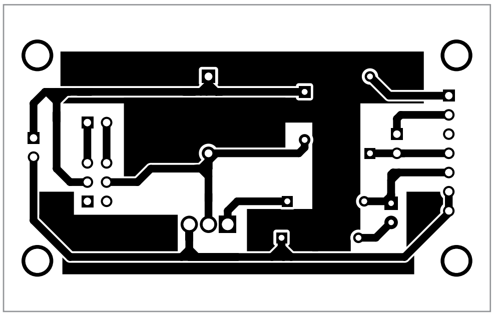

Fig. 7: Actual-size PCB layout for the circuit

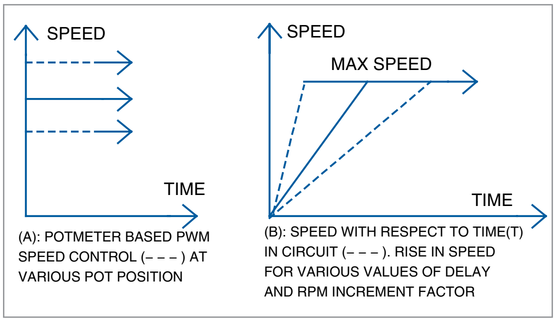

Fig. 9: Graphs representing change in speed of the DC motor

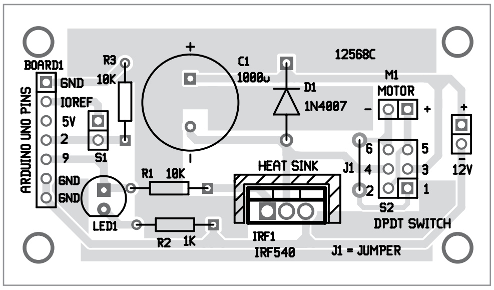

Fig. 8: Component layout of the PCB

Construction and testing

The circuit, including Arduino Uno board and other peripheral components, can be placed in a 10cm x10cm box with a 12V DC jack for power supply input. An actual-size PCB layout suggested for the circuit is shown in Fig. 7 and its component layout in Fig. 8.

Switches S1 and S2 can be placed near the motor at your convenience. LED1 can be on the front panel for visual indication.

Connect a multimeter across pin 9 and GND pin of Arduino to observe numerical incremental voltage on it. This value corresponds to the speed of the DC motor.

The graph shown in Fig. 9(A) represents the change in speed of the motor with respect to change in position of the potentiometer. Graph in Fig. 9(B) represents the change in speed of the DC motor with respect to time after the switch S1 is pressed.

Precautions. Arduino Uno boards are highly sensitive, so handle it carefully. Check the correct input polarity of DC power supply jack (2.1mm centre-positive plug into the board's power jack). Use a suitable heat-sink for MOSFET T1. EFY

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